



Fig. 3: Voltage panel meter with 2V display



Fig. 4: Voltage panel meter showing 3V and 40mA

PARTS LIST

Semiconductors:

- IC1 - LM317 adjustable voltage regulator
 D1-D5 - 1N4007 rectifier diodes
 LED1 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 1-kilo-ohm
 R2 - 680-ohm
 VR1 - 22-kilo-ohm pot

Capacitors:

- C1 - 1000 μ F, 35V electrolytic
 C2 - 100 μ F, 35V electrolytic
 C3 - 0.1 μ F ceramic

Miscellaneous:

- X1 - 230V AC primary to 15V, 1A secondary transformer
 DPM1 - Dual-colour, 3-digit panel display
 S1 - On/off switch
 CON1 - 2-pin power plug
 CON2 - 2-pin connector
 - Heat-sink for LM317

able heat-sink is required for LM317 voltage regulator.

For setting a desired voltage at the output, a resistive voltage divider circuit is used between the output and ground pins of IC1. The voltage divider circuit makes use of fixed resistor R2 (680-ohm) and a pot or variable resistor (22-kilo-ohm). By selecting a correct ratio of feedback resistor (R2) and the variable resistor (VR1), the desired value of

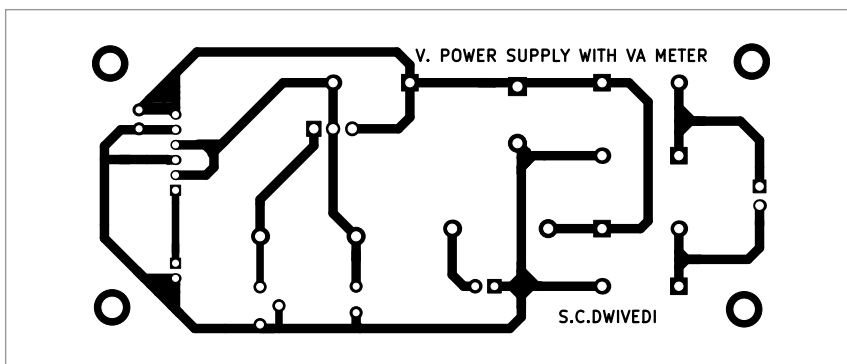


Fig. 5: Actual-size solder-side PCB layout

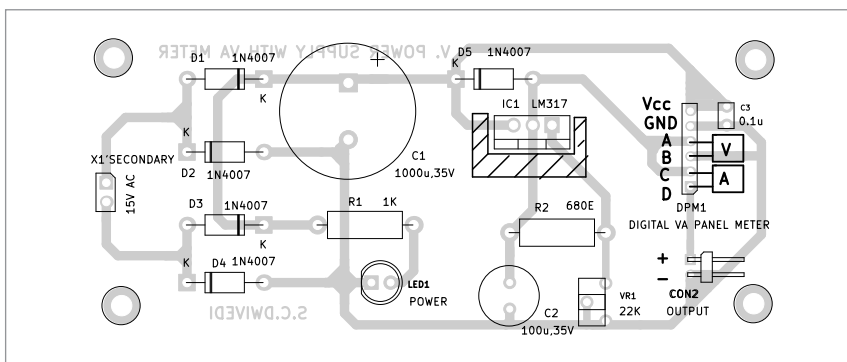


Fig. 6: Component layout of the PCB

output voltage corresponding to the input voltage can be obtained. VR1 may be adjusted from its minimum to maximum position to get 0V to 15V across the load.

When the circuit is switched on, the digital panel meter shows 000 for voltage and 000 for current as well. The panel meter can, however, read voltage up to 100V and current up to 10A. Some important specifications of the panel meter are listed under the table.

Construction and testing

An actual-size, solder-side PCB design for the circuit is shown in Fig. 5 and its component layout is shown in Fig. 6. The circuit, being simple, can also be assembled on a breadboard or general-purpose PCB.

To operate, after checking that the connections have been made properly, switch on the circuit by connecting it to 230V AC mains. To measure voltage across the circuit, connect wires A and B of the panel

meter to points A and B across the load (RL1). The meter will show a voltage reading, which will change as you vary VR1. Assuming that wires C and D are not connected, the reading corresponding to current will be 0.00. The 2V and 0A readings on the panel meter are shown in Fig. 3.

If you want to measure current flowing through the load, connect wires C and D of the panel meter to points C and D in series with the load, as shown in the circuit. The 3V and 0.04 (or 40mA) current readings on the panel meter are shown in Fig. 4.

Depending on the power supply and the load used, you can measure current from 0 to 1.5A. However, the present circuit is designed for voltage up to 15V and current up to 1A only. **EFY**

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